

Week 3 Discussion: Futures (Lec 5–6) Review

Main focus: Futures basics, daily settlement & margins, hedging & basis risk. (Quiz: quick reminders only.)

1 Rapid Review: Lec 5–6 Essentials

1) Futures vs Forward + Close-out P&L

- **Forward (OTC):** typically **no daily settlement**; credit exposure to counterparty; often harder to close out (needs renegotiation/offsetting OTC contract).
- **Futures (exchange + clearinghouse):** **daily settlement (mark-to-market)** via margin; can **close out** anytime by taking the **opposite** position; credit risk is mitigated by margin + clearinghouse.

Close-out P&L over $[t_1, t_2]$:

$$\text{Long futures: P\&L} = F_{t_2} - F_{t_1}, \quad \text{Short futures: P\&L} = F_{t_1} - F_{t_2}.$$

Daily settlement changes the timing of cashflows (realized daily), not the algebraic total P&L over the horizon.

2) Margins & Daily Settlement

Margin account: collateral posted with broker for a futures position.

- **Initial Margin (IM):** deposit when position is opened.
- **Maintenance Margin (MM):** threshold; if balance < MM $\Rightarrow$ **margin call**.
- **Variation Margin:** deposit/withdraw due to daily P&L; after a margin call, deposit to restore balance back to IM (typical convention).

Daily P&L (contract multiplier m):

$$\Delta \text{P\&L}_{\text{long}} = m(F_t - F_{t-1}), \quad \Delta \text{P\&L}_{\text{short}} = m(F_{t-1} - F_t).$$

Margin intuition (order-of-magnitude):

$$\mathbb{E}(|F_j - F_{j-1}|) \ll \text{Initial Margin} \ll F_T.$$

2 Practice Problems

P1. Margin call (silver futures).

Suppose that you enter into a **short** futures contract to sell July silver for \$17.20 per ounce. The size of the contract is 5,000 ounces. The initial margin is \$4,000, and the maintenance margin is \$3,000.

- What change in the futures price will lead to a margin call?
- What happens if you do not meet the margin call?

TA Key for P1

Let ΔF be the **increase** in futures price (\$/oz). A short loses 5,000 ΔF dollars, so

$$\text{Balance} = 4,000 - 5,000 \Delta F.$$

Margin call when balance $< \$3,000$:

$$4,000 - 5,000 \Delta F < 3,000 \implies \Delta F > 0.20.$$

Answer: futures price increase **more than \$0.20/oz**. If margin call not met, broker typically **closes out/liquidates** the position; losses are realized, and any deficit beyond margin must be paid by the trader.

P2. Daily settlement across tax year (crude oil futures).

Suppose that in September 2021 a company takes a **long** position in a contract on May 2022 crude oil futures. It closes out its position in March 2022. The futures price (per barrel) is \$48.30 when it enters into the contract, \$49.10 at the end of December 2021, and \$50.50 when it closes out its position. One contract is for the delivery of 1,000 barrels. The company has a December 31 year end.

- What is the company's **total profit**?
- When** is the profit **realized** (cashflow timing)? *Hint: daily settlement.*

TA Key for P2

Total profit (long futures):

$$(50.50 - 48.30) \times 1,000 = 2.20 \times 1,000 = \$2,200.$$

When realized (daily settlement): gains are realized day-by-day. By Dec 31, 2021, the cumulative gain is

$$(49.10 - 48.30) \times 1,000 = 0.80 \times 1,000 = \$800,$$

and from Jan–Mar 2022 the additional gain is

$$(50.50 - 49.10) \times 1,000 = 1.40 \times 1,000 = \$1,400.$$

So \$800 is realized (cash) during 2021 and \$1,400 during 2022, totaling \$2,200.

Rapid-review checkpoints (common mistakes)

- **Sign check:** long futures gains when F increases; short futures loses when F increases.
- **Daily settlement:** does *not* change total close-out P&L, but it *does* change **when** cash gains/losses hit the margin account.
- **Margin call:** triggered by **margin balance** falling below MM (convert price move $\rightarrow$ dollars using contract size).